

00.

```
use LibraryDB;
```

```
db.libraryBooks.insertMany([
  { "BookID": "B001", "Title": "Introduction to Databases", "Author": "C.J.
Date", "Category": "Database", "Copies": 5, "PublishedYear": 2018, "Status":
"Available" },
  { "BookID": "B002", "Title": "Learning Python", "Author": "Mark Lutz",
"Category": "Programming", "Copies": 2, "PublishedYear": 2019, "Status":
"Borrowed" },
  { "BookID": "B003", "Title": "MongoDB Basics", "Author": "Kristina Chodorow",
"Category": "Database", "Copies": 8, "PublishedYear": 2020, "Status": "Available"
},
  { "BookID": "B004", "Title": "Clean Code", "Author": "Robert C. Martin",
"Category": "Programming", "Copies": 4, "PublishedYear": 2008, "Status":
"Available" },
  { "BookID": "B005", "Title": "Web Development Handbook", "Author": "Jon
Duckett", "Category": "Web", "Copies": 3, "PublishedYear": 2017, "Status":
"Borrowed" },
  { "BookID": "B006", "Title": "JavaScript: The Good Parts", "Author": "Douglas
Crockford", "Category": "Web", "Copies": 7, "PublishedYear": 2009, "Status":
"Available" },
  { "BookID": "B007", "Title": "React Up & Running", "Author": "Stoyan Stefanov",
"Category": "Web", "Copies": 1, "PublishedYear": 2021, "Status": "Borrowed" },
  { "BookID": "B008", "Title": "Advanced Java", "Author": "Herbert Schildt",
"Category": "Programming", "Copies": 6, "PublishedYear": 2016, "Status":
"Available" },
  { "BookID": "B009", "Title": "Cloud Computing Concepts", "Author": "Rajkumar
Buyya", "Category": "Cloud", "Copies": 4, "PublishedYear": 2022, "Status":
"Borrowed" },
  { "BookID": "B010", "Title": "Node.js in Action", "Author": "Mike Cantelon",
"Category": "Programming", "Copies": 3, "PublishedYear": 2015, "Status":
"Available" },
  { "BookID": "B011", "Title": "Fundamentals of AI", "Author": "Stuart Russell",
"Category": "AI", "Copies": 2, "PublishedYear": 2020, "Status": "Available" },
  { "BookID": "B012", "Title": "Machine Learning Basics", "Author": "Tom
Mitchell", "Category": "AI", "Copies": 5, "PublishedYear": 2021, "Status":
"Borrowed" }
]);
```

```
1 //  
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6a0c72fbb148f28f943682d1'),  
    '1': ObjectId('6a0c72fcb148f28f943682d2'),  
    '2': ObjectId('6a0c72fcb148f28f943682d3'),  
    '3': ObjectId('6a0c72fcb148f28f943682d4'),  
    '4': ObjectId('6a0c72fcb148f28f943682d5'),  
    '5': ObjectId('6a0c72fcb148f28f943682d6'),  
    '6': ObjectId('6a0c72fcb148f28f943682d7'),  
    '7': ObjectId('6a0c72fcb148f28f943682d8'),  
    '8': ObjectId('6a0c72fcb148f28f943682d9'),  
    '9': ObjectId('6a0c72fcb148f28f943682da'),  
    '10': ObjectId('6a0c72fcb148f28f943682db'),  
    '11': ObjectId('6a0c72fcb148f28f943682dc')  
  }  
}
```

01.

```
db.libraryBooks.find({ "Category": "Programming" });
```

```
LibraryDB> db.libraryBooks.find({ "Category": "Programming" });
[
  {
    _id: ObjectId('6a0c72fcb148f28f943682d2'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d4'),
    BookID: 'B004',
    Title: 'Clean Code',
    Author: 'Robert C. Martin',
    Category: 'Programming',
    Copies: 4,
    PublishedYear: 2008,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d8'),
    BookID: 'B008',
    Title: 'Advanced Java',
    Author: 'Herbert Schildt',
    Category: 'Programming',
    Copies: 6,
    PublishedYear: 2016,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682da'),
    BookID: 'B010',
    Title: 'Node.js in Action',
    Author: 'Mike Cantelon',
    Category: 'Programming',
    Copies: 3,
    PublishedYear: 2015,
    Status: 'Available'
  },
]
```

02.

```
db.libraryBooks.find({ "Copies": { $Lt: 4 } });
```

```
LibraryDB>
```

```
LibraryDB> db.libraryBooks.find({ "Copies": { $lt: 4 } });
```

```
[
  {
    _id: ObjectId('6a0c72fcb148f28f943682d2'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d5'),
    BookID: 'B005',
    Title: 'Web Development Handbook',
    Author: 'Jon Duckett',
    Category: 'Web',
    Copies: 3,
    PublishedYear: 2017,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d7'),
    BookID: 'B007',
    Title: 'React Up & Running',
    Author: 'Stoyan Stefanov',
    Category: 'Web',
    Copies: 1,
    PublishedYear: 2021,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682da'),
    BookID: 'B010',
    Title: 'Node.js in Action',
    Author: 'Mike Cantelon',
    Category: 'Programming',
    Copies: 3,
    PublishedYear: 2015,
    Status: 'Available'
  }
]
```

03.

```
db.libraryBooks.find({
  "Status": "Available",
  "PublishedYear": { $gt: 2015 }
});
```

```
LibraryDB> db.libraryBooks.find({
|   "Status": "Available",
|   "PublishedYear": { $gt: 2015 }
| });
[
  {
    _id: ObjectId('6a0c72fbb148f28f943682d1'),
    BookID: 'B001',
    Title: 'Introduction to Databases',
    Author: 'C.J. Date',
    Category: 'Database',
    Copies: 5,
    PublishedYear: 2018,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d3'),
    BookID: 'B003',
    Title: 'MongoDB Basics',
    Author: 'Kristina Chodorow',
    Category: 'Database',
    Copies: 8,
    PublishedYear: 2020,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d8'),
    BookID: 'B008',
    Title: 'Advanced Java',
    Author: 'Herbert Schildt',
    Category: 'Programming',
    Copies: 6,
    PublishedYear: 2016,
    Status: 'Available'
  },
  {

```

04.

```
db.libraryBooks.find({ "Title": { $regex: "Basics", $options: "i" } });
```

```
LibraryDB> db.libraryBooks.find({ "Title": { $regex: "Basics", $options: "i" } });
[
  {
    _id: ObjectId('6a0c72fcb148f28f943682d3'),
    BookID: 'B003',
    Title: 'MongoDB Basics',
    Author: 'Kristina Chodorow',
    Category: 'Database',
    Copies: 8,
    PublishedYear: 2020,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682dc'),
    BookID: 'B012',
    Title: 'Machine Learning Basics',
    Author: 'Tom Mitchell',
    Category: 'AI',
    Copies: 5,
    PublishedYear: 2021,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c7612b148f28f943682df'),
    BookID: 'B003',
    Title: 'MongoDB Basics',
    Author: 'Kristina Chodorow',
    Category: 'Database',
    Copies: 8,
    PublishedYear: 2020,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c7612b148f28f943682e8'),
    BookID: 'B012',
    Title: 'Machine Learning Basics',
    Author: 'Tom Mitchell',
    Category: 'AI',
    Copies: 5,
    PublishedYear: 2021,
    Status: 'Borrowed'
  }
]
LibraryDB>
```

05.

```
db.libraryBooks.find({
  $or: [
    { "PublishedYear": { $lt: 2010 } },
    { "Copies": { $gt: 5 } }
  ]
});
```

❏ mongosh mongod://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
LibraryDB> db.libraryBooks.find({
|   $or: [
|     { "PublishedYear": { $lt: 2010 } },
|     { "Copies": { $gt: 5 } }
|   ]
| });
[
  {
    _id: ObjectId('6a0c72fcb148f28f943682d3'),
    BookID: 'B003',
    Title: 'MongoDB Basics',
    Author: 'Kristina Chodorow',
    Category: 'Database',
    Copies: 8,
    PublishedYear: 2020,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d4'),
    BookID: 'B004',
    Title: 'Clean Code',
    Author: 'Robert C. Martin',
    Category: 'Programming',
    Copies: 4,
    PublishedYear: 2008,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d6'),
    BookID: 'B006',
    Title: 'JavaScript: The Good Parts',
    Author: 'Douglas Crockford',
    Category: 'Web',
    Copies: 7,
    PublishedYear: 2009,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d8'),
    BookID: 'B008',
    Title: 'Advanced Java',
    Author: 'Herbert Schildt',
    Category: 'Programming',
    Copies: 6,
  }
]
```

06.

```
db.libraryBooks.find({ "Category": { $ne: "Web" } });
```

```
LibraryDB> db.libraryBooks.find({ "Category": { $ne: "Web" } });
```

```
[
  {
    _id: ObjectId('6a0c72fbb148f28f943682d1'),
    BookID: 'B001',
    Title: 'Introduction to Databases',
    Author: 'C.J. Date',
    Category: 'Database',
    Copies: 5,
    PublishedYear: 2018,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d2'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d3'),
    BookID: 'B003',
    Title: 'MongoDB Basics',
    Author: 'Kristina Chodorow',
    Category: 'Database',
    Copies: 8,
    PublishedYear: 2020,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d4'),
    BookID: 'B004',
    Title: 'Clean Code',
    Author: 'Robert C. Martin',
    Category: 'Programming',
    Copies: 4,
    PublishedYear: 2008,
    Status: 'Available'
  },
]
```



07.

```
db.libraryBooks.find({ "Author": { $in: ["Mark Lutz", "Robert C. Martin"] } });
```

```
LibraryDB> db.libraryBooks.find({ "Author": { $in: ["Mark Lutz", "Robert C. Martin"] } });
[
  {
    _id: ObjectId('6a0c72fcb148f28f943682d2'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d4'),
    BookID: 'B004',
    Title: 'Clean Code',
    Author: 'Robert C. Martin',
    Category: 'Programming',
    Copies: 4,
    PublishedYear: 2008,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c7612b148f28f943682de'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c7612b148f28f943682e0'),
    BookID: 'B004',
    Title: 'Clean Code',
    Author: 'Robert C. Martin',
    Category: 'Programming',
    Copies: 4,
    PublishedYear: 2008,
    Status: 'Available'
  }
]
```

08.

```
db.libraryBooks.find({
  "Status": "Borrowed",
  "Copies": { $lt: 3 }
});
```

```
LibraryDB> db.libraryBooks.find({
|   "Status": "Borrowed",
|   "Copies": { $lt: 3 }
| });
[
  {
    _id: ObjectId('6a0c72fcb148f28f943682d2'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d7'),
    BookID: 'B007',
    Title: 'React Up & Running',
    Author: 'Stoyan Stefanov',
    Category: 'Web',
    Copies: 1,
    PublishedYear: 2021,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c7612b148f28f943682de'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c7612b148f28f943682e3'),
    BookID: 'B007',
    Title: 'React Up & Running',
    Author: 'Stoyan Stefanov',
    Category: 'Web',
  }
]
```

09.

```
db.libraryBooks.find({
  "PublishedYear": { $gte: 2016, $lte: 2020 }
});
```

```
LibraryDB>

LibraryDB> db.libraryBooks.find({
|   "PublishedYear": { $gte: 2016, $lte: 2020 }
| });
[
  {
    _id: ObjectId('6a0c72fbb148f28f943682d1'),
    BookID: 'B001',
    Title: 'Introduction to Databases',
    Author: 'C.J. Date',
    Category: 'Database',
    Copies: 5,
    PublishedYear: 2018,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d2'),
    BookID: 'B002',
    Title: 'Learning Python',
    Author: 'Mark Lutz',
    Category: 'Programming',
    Copies: 2,
    PublishedYear: 2019,
    Status: 'Borrowed'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d3'),
    BookID: 'B003',
    Title: 'MongoDB Basics',
    Author: 'Kristina Chodorow',
    Category: 'Database',
    Copies: 8,
    PublishedYear: 2020,
    Status: 'Available'
  },
  {
    _id: ObjectId('6a0c72fcb148f28f943682d5'),
    BookID: 'B005',
    Title: 'Web Development Handbook',
    Author: 'Jon Duckett',
    Category: 'Web',
    Copies: 3,
    PublishedYear: 2017,
```